Shanmukha Anirudh

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Experience

Clarivate **May 2022**

Software Engineer *Bangalore, Karnataka*

- Designed and developed scalable **Spring Boot microservices** handling **1M+ messages per day** in a **distributed environment**
- Architected **REST APIs to streamline backend communication**, achieving a 15% improvement in **data retrieval efficiency**.
- Built **high-concurrency** backend services **performing heavy file and database I/O with 99.9% service availability**.
- **Migrated** legacy Java 8 **monolith services to Java 21 microservices**, **reducing** response **latency by ~30%**.
- Enforced **asynchronous processing** pipelines using **SQS** to decouple services and **improve throughput.**
- **Created** XSLTs for transformation processing **0.5 million** of XML **records daily**.
- Planned and enforced **Content tracking system** to track individual files across systems which helped to **reduce timeline from 14 days to 3 days**.
- Implemented feature for routing **read-write RDS** instances using **Spring AOP** which helped to **reduce RDS load by 45%**.
- Applied **multithreading** across services for concurrent processing which **reduced processing time by 25%**.
- Wrote test cases using **JUnit with Mockito** with **code coverage of 80%**.
- **Implemented Kafka** for handling task creation for exceptions handles by user which helped system to **store and stream messages**.

Projects

Patient Management System | Springboot, Postgresql, Redis, Kafka, Grpc, Docker

- Built a microservices platform with 6+ independent services.
- Implemented Redis caching reducing database load by ~35%.
- Designed OTP verification using Redis with TTL-based expiry.
- Implemented Redis cache-aside pattern with fallback handling.
- Integrated API Gateway with centralized authentication and rate limiting.
- Applied Circuit Breaker, Retry, Bulkhead, and Timeout strategies reducing failures by ~40%.
- Centralized configuration using Spring Cloud Config Server.

MiniS3 | ***Springboot, Spring security***

- Designed and developed a Mini-S3 object storage system using Java 21 and Spring Boot, supporting bucket creation and file uploads with metadata persistence; implemented file system-based storage capable of handling 10K+ objects per bucket with configurable 50MB file upload limits.
- Refactored the application from a modular monolith to a 3-service microservices architecture (Auth, Bucket, File services), improving service isolation, maintainability, and enabling independent deployment and scaling.
- Implemented secure authentication using Spring Security + JWT (UUID-based identity) and built a reliable upload workflow with validation, size checks, and compensating cleanup logic to prevent orphaned files, ensuring transactional consistency between file system storage and database metadata.

Todo App | ***Angular, Spring Boot, PostgreSQL, Docker***

- Crafted a task management application with features for creating, updating, and deleting tasks, backed by a PostgreSQL database to ensure efficient storage and quick retrieval, enhancing user productivity.
- Utilized Docker for containerization, improving scalability and simplifying deployment across multiple environments.

Technical Skills

Programming Languages: Java, JavaScript, HTML, CSS, XSLT
Framework / Libraries: Spring Boot, Angular, Kafka
Tools: Git, Docker, VS Code, IntelliJ
Databases: MySQL, PostgreSQL
Core Expertise: DSA, OOP, System Design, SDLC, Agile Methodologies, Rest API, Grpc
Distributed Systems: Fault Tolerance, concurrency control
Additional: Debugging, Testing

Achievements

- LeetCode & GeeksforGeeks: 500+ DSA Problems Solved.
- HackerRank: Java (Certified)

Education

Scsvmv University **Aug. 2017 – Apr 2021**
Bachelor of Engineering in Mechanical Engineering CGPA: 9.02 *Kanchi, Tamil Nadu*